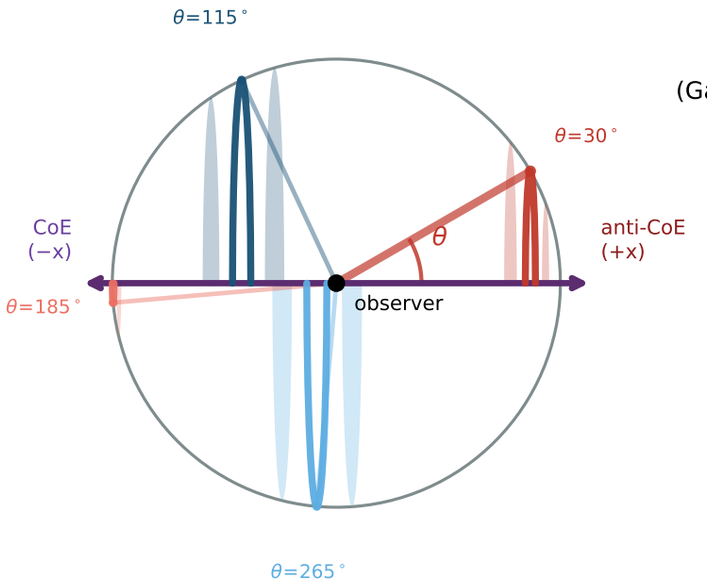
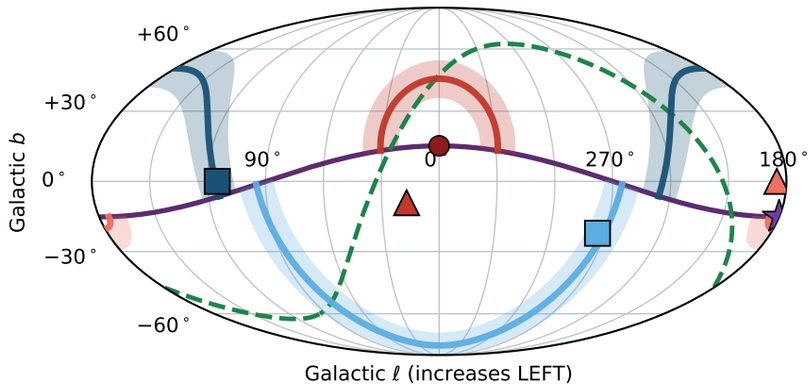
Pure CoE axisymmetry: θ-ray-side half-loops (keep $(n\cdot\hat{y})\sin\theta\geq 0$)
(each 18° end-to-end = $\pm 9^\circ$; MAX $\theta=30^\circ/185^\circ$; MIN $\theta=115^\circ/265^\circ$; anti-CoE $(0^\circ, +15^\circ)$, CoE axis tilted $+15^\circ$)

A. Model frame — θ-ray-side half-loops only
axisymmetric about x = CoE ↔ anti-CoE
keep $(n\cdot\hat{y})\sin\theta\geq 0$ (same side as θ-ray); each 18° end-to-end ($\pm 9^\circ$)

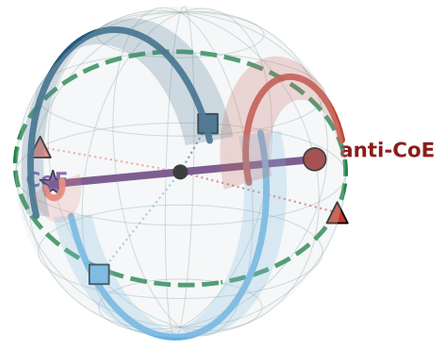


θ-ray-side half-loops · keep $y\,\mathrm{sgn}(\sin\theta)\geq 0$ · each 18° end-to-end ($\pm 9^\circ$)
(after-2ℓ A(ℓ): two MAX + two MIN; open 180° arcs — no full loops)

B. Migkas $A(\ell)$ half-loops on sky (anti-CoE at $(0^\circ, +15^\circ)$)
(Galactic Mollweide; CoE axis tilted $+15^\circ$; θ-ray-side half-loops only; ℓ increases LEFT)



C. Sphere 3D — θ-ray-side half-loops
(open 180° arcs; celestial equator dashed)



MAX $\theta=30^\circ, 185^\circ$; MIN $\theta=115^\circ, 265^\circ$ (θ-ray-side half-loops; 18° end-to-end = $\pm 9^\circ$)

CoE ↔ anti-CoE axis
MAX $\theta=30^\circ$ θ-ray half [21,39]°
MAX $\theta=185^\circ$ θ-ray half [176,194]°
MIN $\theta=115^\circ$ θ-ray half [106,124]°
MIN $\theta=265^\circ$ θ-ray half [256,274]°
celestial equator (Dec = 0)
MAX $(17^\circ, -9^\circ)$
MAX $(185^\circ, +0^\circ)$
MIN $(115^\circ, +0^\circ)$
MIN $(274^\circ, -22^\circ)$
anti-CoE $(0^\circ, +15^\circ)$
CoE $(180^\circ, -15^\circ)$